2. Data Engineering & Ingestion

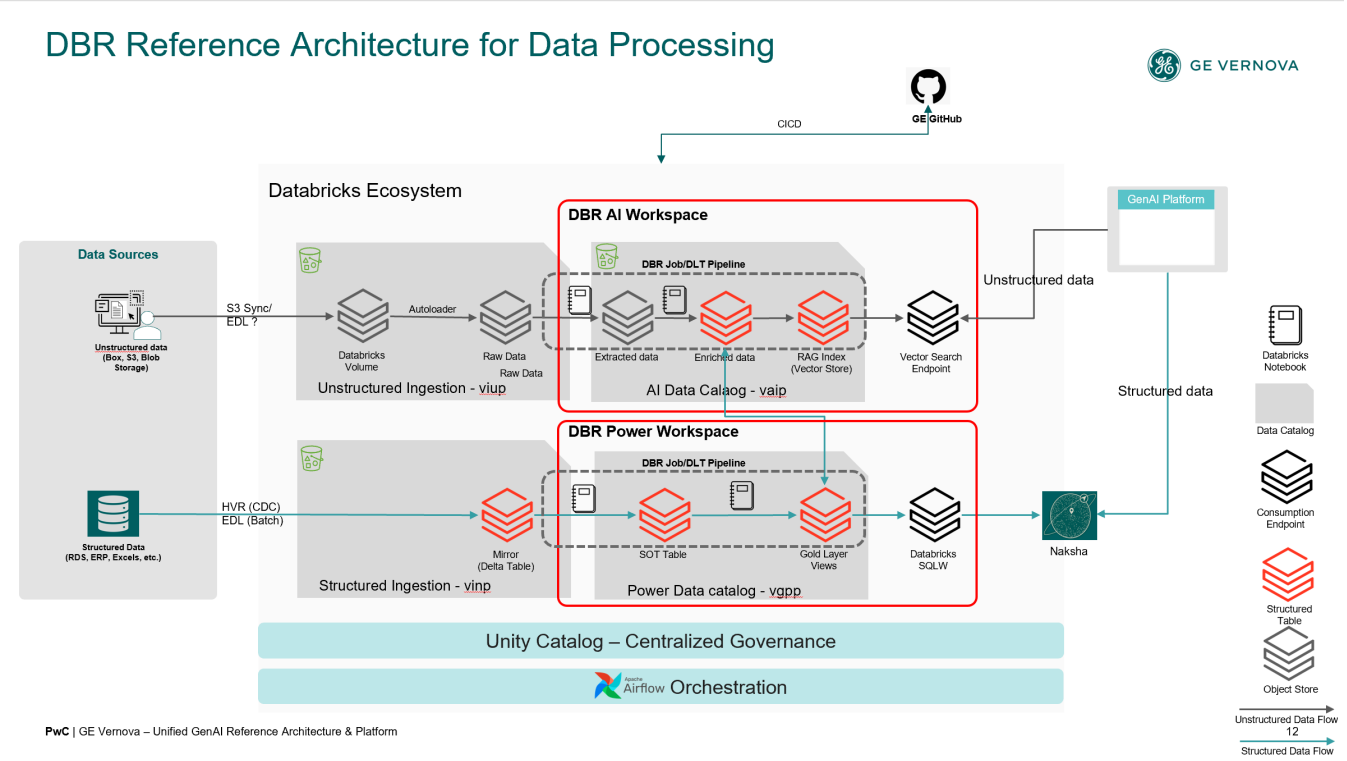
Overview

This section documents the end-to-end data lifecycle for the GenAI Risk Assessment Application. It details how we identify, ingest, normalize, and store the critical fleet data required to power the AI Agent.

The Core Objective: To transform fragmented, multi-format data (PDFs, SQL rows, Word docs) into a unified **Databricks Lakehouse** structure that supports high-speed Vector Search (RAG) and structured SQL querying.

The Databricks Lakehouse Strategy

We utilize a **Medallion Architecture** to organize data as it moves from source to agent:



1. **Bronze Layer (Raw Ingestion):**
    - Exact copies of source data (FSR PDFs, ER JSONs, Milestone Word Docs).
    - *Goal:* Traceability and immutability.
2. **Silver Layer (Cleansed & Enriched):**
    - **OCR Output:** Extracted text from FSRs (Feature F1001.5).
    - **Normalized Metadata:** Dates converted to ISO format (F1001.2), ER numbers standardized.
    - *Goal:* Data quality and searchability.
3. **Gold Layer (Agent-Ready):**
    - **Vector Embeddings:** Chunked text ready for RAG retrieval.
    - **Risk Profiles:** Pre-computed tables for Fleet Heat Maps (F1002).
    - *Goal:* Performance (sub-2 second response times).

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2. Catalog & Workspace Access Control

Access to the above objects is governed via **OneIDM Distribution Lists (DLs)** and Databricks group mappings.

Catalog Access

Catalog	Purpose	Access Distribution Lists
VIUD	Unstructured ingestion (raw documents) for Dev	@GE Vernova VIUD All Looker
VIUP	Processed unstructured datasets for Prod	@GE Vernova VIUP All Looker
VAID	AIready curated datasets for Dev	@GE Vernova VAID Field Service Report Looker @GE Vernova VAID Field Service Report Updater @GE Vernova VAID Field Service Report Consumer
VAIQ	AIready curated datasets for QA	@GE Vernova VAIQ Field Service Report Looker @GE Vernova VAIQ Field Service Report Updater @GE Vernova VAIQ Field Service Report Consumer
VAIP	AI consumption & downstream use for Prod	@GE Vernova VAIP Field Service Report Looker @GE Vernova VAIP Field Service Report Updater @GE Vernova VAIP Field Service Report Consumer

Workspace Access

Environment	Workspace URL	Access Method
Development (Dev)	https://gevernova-ai-dev-dbr.cloud.databricks.com	@GE Vernova VAID SER DBR
QA	https://gevernova-ai-qa-dbr.cloud.databricks.com	@GE Vernova VAIQ SER DBR
Staging(STG)		
Production (Prod)	https://gevernova-ai-dbr.cloud.databricks.com	@GE Vernova VAIP SER DBR
POC	https://gevernova-nrc-workspace.cloud.databricks.com	`@GE Vernova DBR DS Team`

3. Data Catalog & Schema Structure

The project uses a structured data catalog in Databricks Unity Catalog with the following hierarchy:

Structured Data Access

Data Domain	Development (VGPD) DL	Production (VGPP) DL	Description
ER / NCR	@GE Vernova VGPD QLT Consumer	@GE Vernova VGPP QLT Consumer	Engineering Records / NCR data
FSR	@GE Vernova VGPD FSR Consumer	@GE Vernova VGPP FSR Consumer	Field Service Reports
IBAT	@GE Vernova VGPD PRM Consumer	@GE Vernova VGPP PRM Consumer	Installed Base & Asset Tracking

Naksha DL for FSR : @GE Vernova Naksha GP Ser FSR Dev

Unstructured Data Access

Ingestion (VIUD / VIUP)

Catalog	Schema	Object Name	Type	Description
VIUD	`ing_ud_fieldvision`	`fv_field_service_report`	Volume	FieldVision service report documents
VIUD	`ing_ud_spec`	`spec_boroscope_insp_field_service_report`	Volume	Borescope inspection specification documents
VIUD	`ing_ud_fsr_manual`	`manual_field_service_report`	Volume	Manually uploaded Field Service Reports

AI & Consumption (VAID / VAIP)

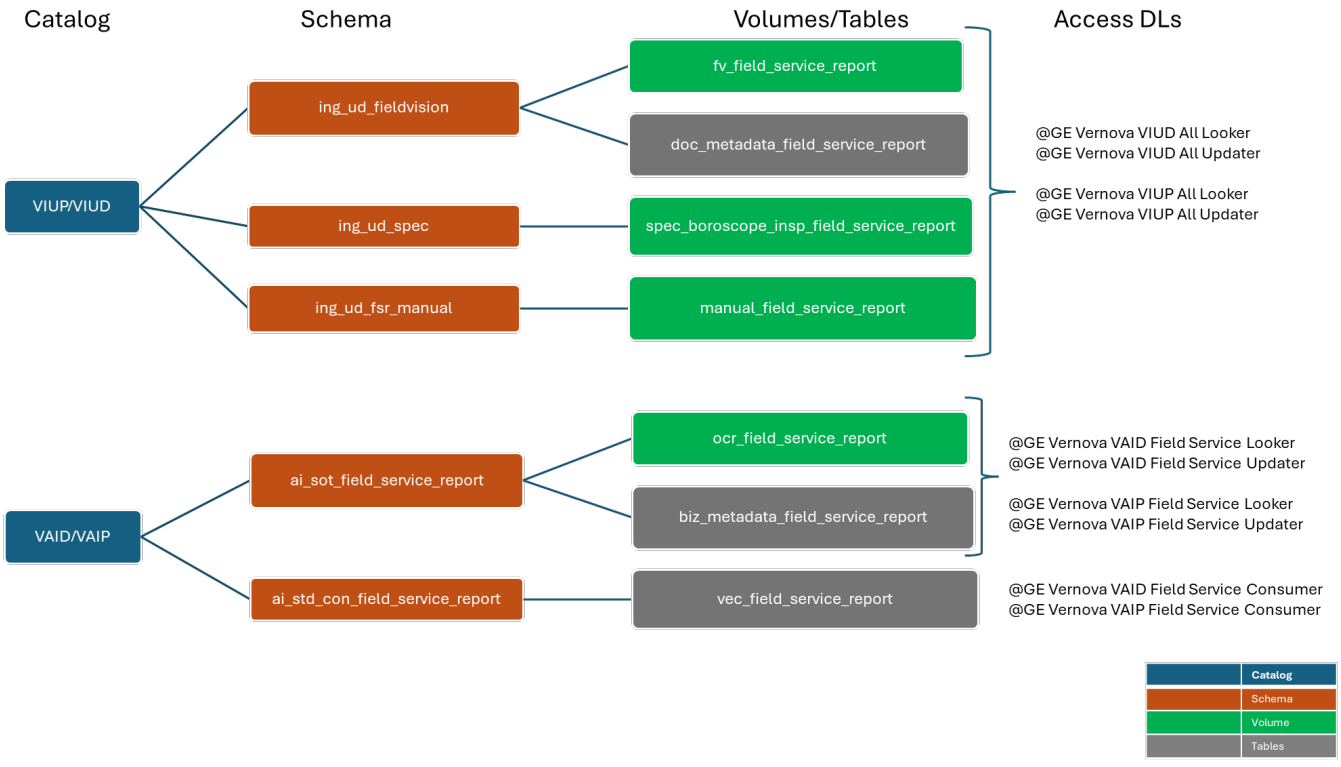
Catalog	Schema	Object Name	Type	Description
VAID	`ai_sot_field_service_report`	`ocr_field_service_report`	Volume	OCRprocessed field service reports
VAID	`ai_sot_field_service_report`	`biz_metadata_field_service_report`	Table	Business metadata extracted from reports
VAID / VAIP	`ai_std_con_field_service_report`	`vec_field_service_report`	Table	Vectorized content for semantic search

Schema Created for SDG

Catalog	Schema	Name of the volume	Path Volume
VAID	ai_sot_field_service_report	ocr_feild_service_report	s3://ge-vaid/ocr_field_service_report/ocr_field_service_report
	ai_std_con_field_service_report		
VAIQ	ai_sot_field_service_report	ocr_field_service_report	s3://ge-vaiq/ai_sot_field_service_report/ocr_field_service_report
	ai_std_con_field_service_report		

Ingestion Schema

Catalog	Schema	Name of the volume	Path Volume
VIUD	ing_ud_fieldvision	fv_field_service_report	s3://ge-viud/fv_field_service_report
	ing_ud_fsr_manual	manual_field_service_report	s3://ge-viud/manual_field_sercive_report
	ing_ud_spec	spec_boroscope_insp_field_servive_report	s3://ge-viud/spec_boroscope_insp_field_servive_report



4.. How to Join a Distribution List (DL)

To request access to a required DL:

1. Navigate to the OneIDM Distribution List portal:
   https://oneidm.ge.com/faces/modules/my_groups/distribution_lists_join.xhtml
2. Search for the required Distribution List by name.
3. Select the DL.

4. Click **Join** and complete the request.

> ***Note:*** *Approval may be required depending on the DL and data sensitivity.*

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5. Environment Details

Dev Environment Details

Category	Item	Name / Details	Notes for End Users
Databricks Workspace	Workspace URL	https://gevernova-ai-dev-dbr.cloud.databricks.com	Primary Dev workspace for AI
	AllPurpose Cluster	`ai-pw-ser-ds-dev-apc`	Use for development of notebooks and code
	SQL Warehouse	`ai-pw-ser-ds-dev-sqlw`	Use for BI & SQL analytics, to ren queries
Source Control	GitHub Repository	`ai_pw_ser_dbr`	To store project related notebooks, files , jobs ,etc.
	Repository URL	https://github.apps.gevernova.net/GEV-SoX-DataBricks/ai_pw_ser_dbr.git	GitHub Enterprise access required
AWS Storage	AI Catalog S3 Bucket	`ge-vaid`	Backing storage for AI data for processing and consumption
	Unstructured Ingestion S3 Bucket	`ge-viud`	Common ingestion catalog for unstructured data
Unity Catalog	Catalog	`VAID`	To store AI related datasets
	Catalog	`VIUD`	To store Unstructured ingestion datasets
	CrossCatalog Access	`VGPD`, `VGPP`	Extsing catalog housing consumption ready structured data for Power
VAID Schema	Schema	`ai_sot_field_service_report`	OCR and extracted content
	Volume	`ocr_field_service_report`	OCR output files
	S3 Path	`s3://ge-vaid/ocr_field_service_report`	Access via Unity Catalog only
	Schema	`ai_std_con_field_service_report`	To house final consumption tables and vector search indexes.
VIUD Schema	Schema	`ing_ud_fieldvision`	FieldVision ingestion Schema for unstructured data
	Volume	`fv_field_service_report`	To FSR PDFs coming from Field Vision
	S3 Path	`s3://ge-viud/fv_field_service_report`	Readonly for most users
	Schema	`ing_ud_fsr_manual`	Manual FSR uploads
	Volume	`manual_field_service_report`	Volume to store the actual files
	S3 Path	`s3://ge-viud/manual_field_service_report`	Readonly for most users
	Schema	`ing_ud_spec`	SPEC ingestion Schema for unstructured data
	Volume	`spec_boroscope_insp_field_service_report`	To house Borescope inspection PDF reports
	S3 Path	`s3://ge-viud/spec_boroscope_insp_field_service_report`	Readonly for most users

QA Environment Details

Category	Item	Name / Details	Notes for End Users
Databricks Workspace	Workspace URL	https://gevernova-ai-qa-dbr.cloud.databricks.com/	Primary Dev workspace for AI
	AllPurpose Cluster	`ai-pw-ser-ds-qa-apc`	Use for development of notebooks and code
	SQL Warehouse	ai-pw-ser-ds-qa-sqlw	Use for BI & SQL analytics, to ren queries
Source Control	GitHub Repository	`ai_pw_ser_dbr`	To store project related notebooks, files , jobs ,etc.
	Repository URL	https://github.apps.gevernova.net/GEV-SoX-DataBricks/ai_pw_ser_dbr.git Branch: QA	GitHub Enterprise access required
AWS Storage	AI Catalog S3 Bucket	`ge-vaiq`	Backing storage for AI data for processing and consumption

	Unstructured Ingestion S3 Bucket	`ge-viud`	Common ingestion catalog for unstructured data
Unity Catalog	Catalog	`VAIQ`	To store AI related datasets
	CrossCatalog Access	`VGPD`, `VGPP`	Extsing catalog housing consumption ready structured data for Power
VAIQ Schema	Schema	`ai_sot_field_service_report`	OCR and extracted content
	Volume	`ocr_field_service_report`	OCR output files
	S3 Path	`s3://ge-vaid/ocr_field_service_report`	Access via Unity Catalog only
	Schema	`ai_std_con_field_service_report`	To house final consumption tables and vector search indexes.

POC Environment Details

The following resources are used for the Proof of Concept (POC) environment:

Component Type	Name / Details
Workspace URL	https://gevernova-nrc-workspace.cloud.databricks.com/
All-Purpose Cluster	`dev-cluster`
SQL Warehouse	`dev-ml-sql`
Compute Policy	`nrc-job-policy`
Vector Search Service	`nrc-vector-search`
Distribution List	`@GE Vernova DBR DS Team`

Consumption

Structured data

The LLM Agent is expected to consume the stetcured data from databricks via Naksha Platfrom. Naksha Intgeration documnetation can be found here.

Unscturetured data

For unstctured data the consumption will happen via Vector Search Endpoint. The LLM Agent is expected to query the Vector Serach Indices using standard Databricks SDK. Prease refer to the official documenatation here.

Data Source Inventory

The following systems are in-scope for ingestion (Refer to **Feature F1014**).

Source System	Data Type	Ingestion Method	Key Use Case
Event Reports (ER DB)	Structured SQL	Batch JDBC	Historical risk drivers & symptoms.
FieldVision / Box	Unstructured (PDF)	API / Cloud Storage	Field Service Reports (FSRs) & visual findings.
IBAT	Structured	API	Unit attributes (Cooling type, Config).
PRISM	Structured	SQL Connect	Quantitative risk scores & probability.
Event Vision	Structured	API	Outage schedules & prioritization.
OSA Tool	Semi-Structured (DOCX)	File Ingestion	Milestone context & previous recommendations.
Process Docs	Documents (PDF)	Cloud Storage	Technical Manuals (TM), GEK, TILs for grounding.

Documentation Tree

Select a topic below for detailed technical specifications.

2.1 Data Ingestion Strategy

- **Feature Reference:** F1014
- **Content:** Details on Databricks Workflows, pipeline scheduling (Batch vs. Real-time), and connection security for each source system.

2.2 OCR & Document Parsing

- **Feature Reference:** F1001.5 (FSRs), F1020 (Milestone Docs)
- **Content:** The technical logic for converting **image-based PDF tables** in FSRs into machine-readable text.
- **Milestones:** How Word documents from OSA are parsed to extract scope and deliverables.

2.3 Data Schema & Models

- **Feature Reference:** F1010 (Risk Schema), F1001.7 (Component Scoping)
- **Content:** The **Standardized 5-Part Risk Schema** (Equipment Component Context Risk Factor Evidence).
- **Metadata:** How we tag data to ensure a "Stator" query never returns "Rotor" data.

Data Governance Principles

These principles must be adhered to across all ingestion pipelines:

1. **Traceability First:** Every single row or vector embedding must retain metadata pointing to its origin (e.g., `Source: FSR-123.pdf, Page 4`). This is critical for **Feature F1001.11 (Precise Citations)**.
2. **No Hallucinations via Filtering:** Ingestion pipelines must rigorously tag data by **Component Type** (Stator vs. Rotor) to prevent cross-contamination during retrieval (Feature F1001.7).
3. **Missing Data Awareness:** Pipelines must identify when expected data (e.g., an FSR for a completed outage) is missing and flag it for the **Feedback Loop** (Feature F1004.2).

Key Contacts

- **Data Engineering POD Lead:** *[Insert Name]*
- **D&A Solution Architect:** *[Insert Name]*
- **Databricks Platform Admin:** *[Insert Name]*